Notes and Selected Exercises for: Precalculus, Stitz, Zeager

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Chapter 1

Relations and Functions

1.1 Sets of Real Numbers and the Cartesian Coordinate Plane

1.1.1 Sets of Numbers

Definition 1.1.1 (Set)**.** A **set** is a well-defined collection of objects which are called the 'elements' of the set. Here, 'well-defined' means that it is possible to determine if something belongs to the collection or not, without prejudice.

Definition 1.1.2. Ways to describe sets

1. **The Verbal Method**: Use a sentence to define a set.
2. **The Roster Method**: Begin with a left brace '{', list each element of the set only once and then end with a right brace '}'.
3. **The Set-Builder Method**: A combination of the verbal and roster methods using a "dummy variable" such as x.

Definition 1.1.3. Sets of Numbers

1. The **Empty Set**: $\emptyset = \{\} = \{x \mid x \neq x\}$. This is the set with no elements.
2. The **Natural Numbers**: $\mathbb{N} = \{1, 2, 3, ...\}$. The periods of ellipsis here indicate that the natural numbers contain 1, 2, 3, 'and so forth'.
3. The **Whole Numbers**: $\mathbb{W} = \{0, 1, 2, ...\}$
4. The **Integers**: $\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$
5. The **Rational Numbers**: $\mathbb{Q} = \{\frac{a}{b} \mid a \in \mathbb{Z} \wedge b \in \mathbb{Z}\}$. Rational numbers are the ratios of integers.
6. The **Real Numbers**: $\mathbb{R} = \{x \mid x \text{ possesses a decimal representation.}\}$

7. The **Irrational Numbers**: $\mathbb{P} = \{x \mid x \text{ is a non-rational real number.}\}$ Said another way, an irrational number is a decimal which neither repreats nor terminates.

8. The **Complex Numbers**: $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R} \wedge i = \sqrt{-1}\}$.

Definition 1.1.4 (Intervals)**.** Visualizing $\mathbb{R}$ as a line, segments on this line are called **intervals** of numbers. An interval can be described as a set in the form $\{x \mid a < x < b\}$, or in interval notation (a, b)

Definition 1.1.5 (Set Operations)**.** Supposing that A and B are two sets.

- The **intersection** of A and B: $A \cap B = \{x \mid x \in A \wedge x \in B\}$
- The **union** of A and B: $A \cup B = \{x \mid x \in A \vee x \in B\}$

Definition 1.1.6 (Cartesian Coordinate System)**.** The **Cartesian Coordinate System** is a coordinate system that specifies each point on it uniquely by a pair of real numbers called coordinates. In a Cartesian Coordinate Plane:

- (a, b) and (c, d) represent the same point in the plane if and only if $a = c \wedge b = d$
- (x, y) lies on the *x-axis* if and only if $y = 0$
- (x, y) lies on the *y-axis* if and only if $x = 0$
- The origin is the point $(0, 0)$. It is the only point common to both axes.

Definition 1.1.7 (Symmmetry)**.** Two points (a, b) and (c, d) in the plane are said to be:

- **symmetric about the *x-axis*** if $a = c \wedge b = -d$
- **symmetric about the *y-axis*** if $a = -c \wedge b = d$
- **symmetric about the *origin*** if $a = -c \wedge b = -d$

Definition 1.1.8 (Reflections)**.** To reflect a point (x, y) about the:

- *x-axis*, replace y with $-y$
- *y-axis*, replace x with $-x$
- *origin*, replace x with $-x$ and y with $-y$

Definition 1.1.9 (Distance)**.** Supposing we hace two points $P(x_0, y_0)$ and $Q(x_1, y_1)$ in the plane. The **distance** between P and Q is the length of the line segment joining P with Q. Using the pythagorean theorem we can derive the distance formula as:

$$d = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2}$$

Definition 1.1.10. The midpoint M of the line segment connecting $P(x_0, y_0)$ and $Q(x_1, y_1)$ is:

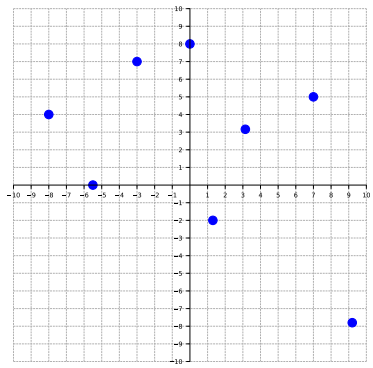
$$M = (\frac{x_0 + x_1}{2}, \frac{y_0 + y_1}{2})$$

1.1.2 Problems

2. $(-1, 5] \cap [0, 8) = [0, 5]$

8. $(-\infty, 5) \cup (5, \infty)$

20.



22.

$$d = \sqrt{(-3-1)^2 + (5-2)^2} = \sqrt{16+9} = \sqrt{25} = 5$$
$$M = (\frac{1-3}{2}, \frac{2+5}{2}) = (-1, 3.5)$$

30.

$$4 = \sqrt{(3-x)^2 + (2+1)^2} = \sqrt{9 - 6x + x^2 + 9} = \sqrt{x^2 - 6x + 18}$$
$$x^2 - 6x + 18 = 16$$
$$x^2 - 6x + 2 = 0$$
$$x_1 = 3 + \sqrt{7}, x_2 = 3 - \sqrt{7}$$
$$(3 + \sqrt{7}, -1), (3 - \sqrt{7}, -1)$$

34. He's at $(-3, -4)$, at a distance of 5 miles, after running he will be at $(4, -4)$

35. (a)
$$d = \sqrt{(a-a)^2 + (y_1 - y_0)^2} = \sqrt{(y_1 - y_0)^2} = y_1 - y_0$$

(b)
$$d = \sqrt{(x_1 - x_0)^2 + (b-b)^2} = \sqrt{(x_1 - x_0)^2} = x_1 - x_0$$

(c)
$$d = \sqrt{(x-x)^2 + (y-y)^2} = \sqrt{0} = 0$$

36.

$$M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$$

$$\begin{aligned}
d_{PM} &= \sqrt{(\frac{x_1 + x_2}{2} - x_1)^2 + (\frac{y_1 + y_2}{2} - y_1)^2} \\
&= \sqrt{(\frac{x_2 - x_1}{2})^2 + (\frac{y_2 - y_1}{2})^2} \\
&= \sqrt{\frac{x_1^2 - 2x_1x_2 + x_2^2}{4} + \frac{y_1^2 - 2y_1y_2 + y_2^2}{4}} \\
d_{QM} &= \sqrt{(\frac{x_1 + x_2}{2} - x_2)^2 + (\frac{y_1 + y_2}{2} - y_2)^2} \\
&= \sqrt{(\frac{x_1 - x_2}{2})^2 + (\frac{y_1 - y_2}{2})^2} \\
&= \sqrt{\frac{x_1^2 - 2x_1x_2 + x_2^2}{4} + \frac{y_1^2 - 2y_1y_2 + y_2^2}{4}}
\end{aligned}$$

$$d_{PM} = d_{QM}$$

37. To prove that the points are the vertices of a right triangle there must be an internal angle with a value of 90deg or prove the converse of the pythagorean theorem.

    (a)

    $$AB = \sqrt{13}, AC = 2\sqrt{13}, BC = \sqrt{65}$$
    $$BC^2 = 65 = AB^2 + AC^2 = 13 + 52$$

    (b)

    $$AB = 5\sqrt{2}, AC = 5, BC = 5$$
    $$AB^2 = 50 = AC^2 + BC^2 = 25 + 25$$

38. To find the remaining point that makes the square, we must find a point that is at the same distance as the other points. We can create a vector from one of the sides of the square and just use it as a base to create the square. If we take $CB = (4, 3)$, then applying that to A, we get $D = (1, 4)$

1.2 Relations

Definition 1.2.1 (Relation)**.** A **relation** is a set of points in the plane.

Definition 1.2.2 (The Fundamental Graphing Principle)**.** Supposing there exists a relation R. Plotting the points of the set R in the plane, produces the **graph** of R

Definition 1.2.3 (The Fundamental Graphing Principle)**.** The graph of an equation is the set of points which satisfices the equation that describes the relation. That is, a point (x, y) is on the graph of the equation if and only if x and y satisfy the equation.

Definition 1.2.4 (Intercepts)**.** The places where a graph crosses or touches the axes are called **intercepts**, intercepts come in two distinct varieties:

- A point on a graph which is also on the *x-axis* is called an **x-intercept** of the graph
- A point on a graph which is also on the *y-axis* is called an **y-intercept** of the graph

Definition 1.2.5 (Finding the Intercepts of the Graph of an Equation)**.** Given an equation involving x and y, we find the intercepts of the graph as follows:

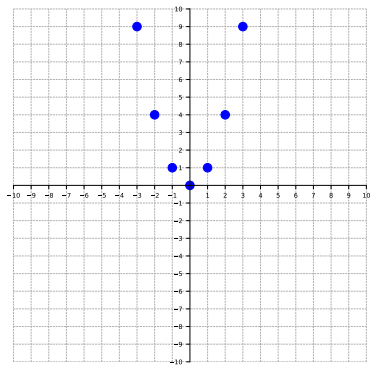
- *x-intercepts* have the form $(x, 0)$; set $y = 0$ in the equation and solve for x
- *y-intercepts* have the form $(0, y)$; set $x = 0$ in the equation and solve for y

Definition 1.2.6 (Testing the Graph of an Equation for Symmetry)**.** To test the graph of an equation for Symmetry

- about the *y-axis* - substitute $(-x, y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the *y-axis*.
- about the *x-axis* - substitute $(x, -y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the *x-axis*.
- about the *origin* - substitute $(-x, -y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the *origin*.

1.2.1 Problems

1.



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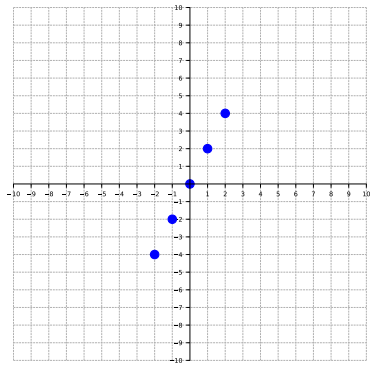

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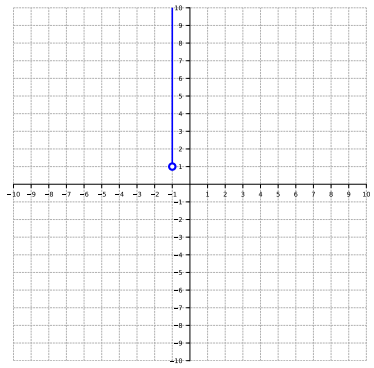


21.

$$\{(-4,-1),(-2,1),(0,3),(1,4)\} = \{(m, m+3) \mid m \in \{-4,-2,0,1\}\}$$

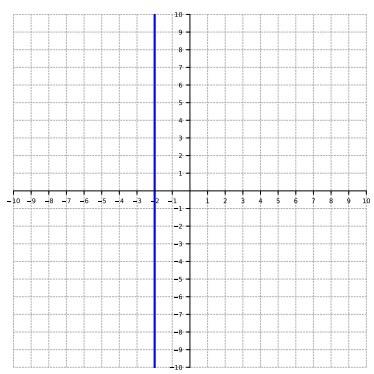
23.

$$\{(2,y) \mid y > -3\}$$

28.

$$\{(x,y) \mid -3 < x \leq 2\}$$

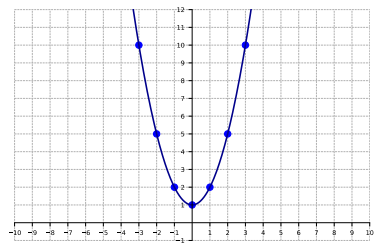
31.



37. We can start by plotting all the points for which y = 0, that is all the odd integers, then we just copy those points and translate them into the even integers on y. Will look like a grid pattern. Points in the relation: $(1,0),(3,2)$, Points not in the relation $(1,1),(2,2)$

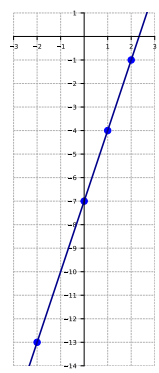
39. Practically the it is the graph of the equation $y = 2^x$

41.
- *x-intercept* doesn't exists, *y-intercept* at 0
-



- Graph is symmetric about the *y-axis*

47.
- *x-intercept* at $\frac{7}{3}$, *y-intercept* at -7
-



- Graph is not symmetric, example using $-x$ we see that $y = 3x - 7 \neq -3x - 7$

53. Multiple values of x can be multiple values of y and viceversa

1.3 Introduction to Functions

Definition 1.3.1 (Function)**.** A relation in which each *x-coordinate* is matched with only one *y-coordinate* is said to describe y as a **function** of x

Definition 1.3.2 (Vertical Line Test)**.** A set of points in the plane represents a y as a function of x if and only if no two points lie on the same vertical line.

Definition 1.3.3 (Domain and Range of a function)**.** Suppose that F is a relation which describes y as a function of x

- The set of the *x-coordinates* of the points in F is called the **domain** of F
- The set of the *y-coordinates* of the points in F is called the **range** of F

1.3.1 Problems

1. Function, domain: $\{-3, -2, ..., 2, 3\}$, range: $\{0, 1, 4, 9\}$

5. Not a function, example $(1, 2)$ and $(1, 4)$

11. Function, domain $(-\infty, \infty)$, range $[0, \infty)$

13. Function, domain $[-4, 1]$, range $[-1, 4]$

17. Function, domain $[2, \infty)$, range $[0, \infty)$

21. Function, domain $[-2, \infty)$, range $[-3, \infty)$

33. y is a function of x

47. y is not a function of x, example for $x = 1$, we have that $y = 1$ or $y = -1$

48. Sasquatches might enter or leave the area, Sasquatches can be born or they can die, range would be $[0, \infty)$

49. Some of them might have multiple emails. Phone numbers would also not be a function since they can also have multiple phone numbers. Social security numbers is unique since it is supposed to be unique.

50. Find the roots of the equation.

1.4 Function Notation

Definition 1.4.1 (Function Notation)**.** If we think of the domain of a function as a set of **inputs** and the range as a set of **outputs**, we can think of a function f as a **process** by which each input x is matches with only one output y. Since the function is completely determined by the input x and thr process f, we symbolize the output with **function notation**: $f(x)$, read 'f of x'. The value of y is completely dependent on the choice of x. For this reason, x is often called the **independent variable**, or **argument** of f, whereas y is often called the **dependent variable**.

1.4.1 Problems

1. $f(x) = \frac{2x+3}{4}$, domain $(-\infty, \infty)$

10. $f(x) = \sqrt{\frac{4}{x}} - 13$, domain $(0, \infty)$

12. (a) $f(3) = 3 - 4(3) = 3 - 12 = -9$
    (b) $f(-1) = 3 - 4(-1) = 3 + 4 = 7$
    (c) $f(\frac{3}{2}) = 3 - 4(\frac{3}{2}) = 3 - 6 = -3$
    (d) $f(4x) = 3 - 4(4x) = 3 - 16x$
    (e) $4f(x) = 4(3 - 4x) = 12 - 16x$
    (f) $f(-x) = 3 + 4x$
    (g) $f(x - 4) = 3 - 4(x - 4) = 3 - 4x + 16 = 19 - 4x$
    (h) $f(x) - 4 = 3 - 4x - 4 = -4x - 1$
    (i) $f(x^2) = 3 - 4x^2$

33.
$$f(0) = \frac{3}{4-0} = \frac{3}{4}$$
$$f(x) \text{ is never equal to } 0$$

37. $(-\infty, \infty)$

63. $A(3) = 9$, $x^2 = 36; x = \pm\sqrt{36} = \pm 6$

1.5 Function Arithmetic

Definition 1.5.1 (Function Arithmetic)**.** Suppose that f and g are functions and x is in both the domain of f and the domain of g.

- The **sum** of f and g, denoted $f+g$, is the function defined by the formula
$$(f + g)(x) = f(x) + g(x)$$
- The **difference** of f and g, denoted $f - g$, is the function defined by the formula
$$(f - g)(x) = f(x) - g(x)$$
- The **product** of f and g, denoted fg, is the function defined by the formula
$$(fg)(x) = f(x)g(x)$$
- The **quotient** of f and g, denoted $\frac{f}{g}$, is the function defined by the formula
$$(\frac{f}{g})(x) = \frac{f(x)}{g(x)}$$

Definition 1.5.2 (Difference Quotient)**.** Given a function f, the **difference quotient** of f is the expression

$$\frac{f(x+h)-f(x)}{h}$$

1.5.1 Problems

1.

$$(f+g)(2) = f(2)+g(2) = 3(2)+1+4-2 = 6+3 = 9$$
$$(f-g)(-1) = f(-1)-g(-1) = 3(-1)+1-4+(-1) = -3+1-4-1 = -7$$
$$(g-f)(1) = g(1)-f(1) = 4-1-3(1)-1 = -1$$
$$(fg)(\frac{1}{2}) = (3(\frac{1}{2})+1)(4-\frac{1}{2}) = (\frac{5}{2})(\frac{7}{2}) = \frac{35}{4}$$
$$(\frac{f}{g})(0) = \frac{1}{4}$$
$$(\frac{g}{f})(-2) = \frac{4-(-2)}{3(-2)+1} = -\frac{6}{5}$$

11.
- Domain $(-\infty,\infty)$, $(f+g)(x) = 3x-1$
- Domain $(-\infty,\infty)$, $(f-g)(x) = x+3$
- Domain $(-\infty,\infty)$, $(fg)(x) = 2x^2-4x+x-2 = 2x^2-3x-2$
- Domain $(-\infty,2)\cup(2,\infty)$, $(\frac{f}{g})(x) = \frac{2x+1}{x-2}$

21.

$$f(x) = 2x-5; \frac{f(x+h)-f(x)}{h} = \frac{2(x+h)-5-2x+5}{h} = \frac{2h}{h} = 2$$

51.

$$(f+g)(-3) = 4-2 = 2$$

1.6 Graphs of Functions

Definition 1.6.1 (The Fundamental Graphing Principle for Functions)**.** The graph of a function f is the set of points which satisfy the equation $y = f(x)$. That is, the point (x,y) is on the graph if and only if $y = f(x)$

Definition 1.6.2 (Zeros of a function)**.** The **Zeros** of a function f are the solutions to the equation $f(x) = 0$. In other words, x is a zero of f if and only if $(x,0)$ is an x-intercept of the graph of $y = f(x)$

Definition 1.6.3 (Testing the Graph of a Function for Symmetry)**.** The graph of a function f is symmetric

- about the y-axis if and only if $f(-x) = f(x)$ for all x in the domain of f
- about the origin if and only if $f(-x) = -f(x)$ for all x in the domain of f

1.6.1 General Function Behavior

Definition 1.6.4 (Increasing, Decreasing and Constant Functions)**.** Suppose f is a function defined on an interval I. We say f is:

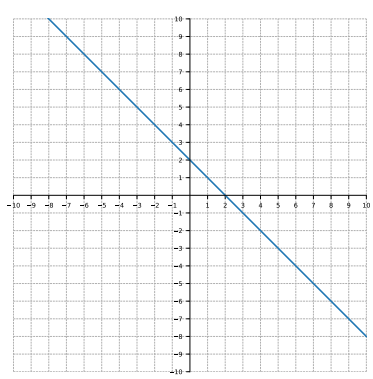
- **increasing** on I if and only if $f(a) < f(b)$ for all real numbers $a, b \in I(a < b)$
- **decreasing** on I if and only if $f(a) > f(b)$ for all real numbers $a, b \in I(a < b)$
- **constant** on I if and only if $f(a) = f(b)$ for all real numbers $a, b \in I$

Definition 1.6.5 (Maximums and Minimums of a Function)**.** Suppose that f is a function with $f(a) = b$

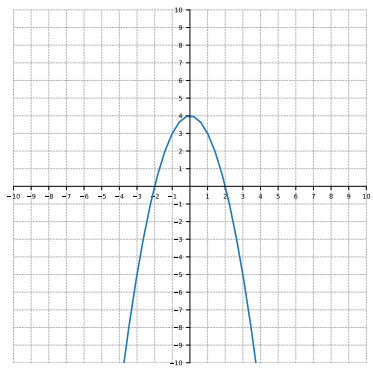
- We say f has a **local maximum** at the point (a, b) if and only if there is an open interval I containing a for which $f(a) \geq f(x)$ for all $x \in I$. The value $f(a) = b$ is called 'a local maximum value of f' in this case.
- We say f has a **local minimum** at the point (a, b) if and only if there is an open interval I containing a for which $f(a) \leq f(x)$ for all $x \in I$. The value $f(a) = b$ is called 'a local minimum value of f' in this case.
- The value b is called the **maximum** of f if $b \geq f(x)$ for all x in the domain of f
- The value b is called the **minimum** of f if $b \leq f(x)$ for all x in the domain of f

1.6.2 Problems

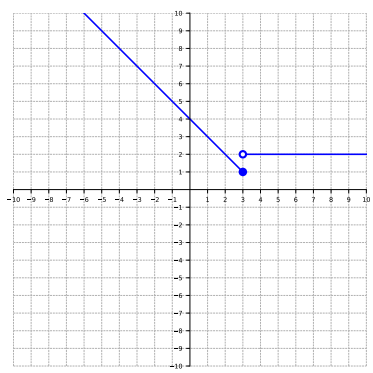
1. Domain $(-\infty, \infty)$, intercepts: $(2, 0)$, Not symmetric



4. Domain $(-\infty, \infty)$, intercepts: $(-2, 0), (2, 0)$, Symmetric about the y-axis



13.



21. Neither even nor odd.

$$f(x) = 7x \neq -7x = f(-x); f(x) = 7x \neq -7x = -f(x)$$

28. Function is even

$$f(x) = 4 - x^2 = f(-x); f(x) = 4 - x^2 \neq= x^2 - 4 = -f(x)$$

1.7 Transformations

Definition 1.7.1 (Vertical Shifts)**.** Suppose f is a function and k is a positive number.

- To graph $y = f(x) + k$, shift the graph of $y = f(x)$ up k units by adding k to the y-coordinates of the points on the graph of f
- To graph $y = f(x) + k$, shift the graph of $y = f(x)$ down k units by subtracting k to the y-coordinates of the points on the graph of f

Definition 1.7.2 (Horizontal Shifts)**.** Suppose f is a function and h is a positive number.

- To graph $y = f(x + h)$, shift the graph of $y = f(x)$ left h units by subtracting h to the x-coordinates of the points on the graph of f
- To graph $y = f(x - h)$, shift the graph of $y = f(x)$ right h units by subtracting h to the x-coordinates of the points on the graph of f

Definition 1.7.3 (Reflections)**.** Suppose f is a function

- To graph $y = -f(x)$, reflect the graph of $y = f(x)$ across the x-axis by multiplying the y-coordinates of the points on the graph of f by -1
- To graph $y = f(-x)$, reflect the graph of $y = f(x)$ across the y-axis by multiplying the x-coordinates of the points on the graph of f by -1

Definition 1.7.4 (Vertical Scalings)**.** Suppose f is a function and $a > 0$. To graph $y = af(x)$, multiply all of the y-coordinates of the points on the graph of f by a. We say the graph of f has been vertically scaled by a factor of a

- If $a > 1$, we say the graph of f has undergone a vertical stretching (expansion, dilation) by a factor of a.
- If $0 < a < 1$, we say the graph of f has undergone a vertical shrinking (compresion, contraction) by a factor of $\frac{1}{a}$

Definition 1.7.5 (Horizontal Scalings)**.** Suppose f is a function and $b > 0$. To graph $y = f(bx)$, divide all of the x-coordinates of the points on the graph of f by b. We say the graph of f has been horizontally scaled by a factor of $\frac{1}{a}$

- If $0 < b < 1$, we say the graph of f has undergone a horizontal stretching (expansion, dilation) by a factor of $\frac{1}{b}$
- If $b > 1$, we say the graph of f has undergone a horizontal shrinking (compresion, contraction) by a factor of b.

Definition 1.7.6 (Transformations)**.** Suppose f is a function. If $A \neq 0$ and $B \neq 0$, then to graph

$$g(x) = Af(Bx + H) + K$$

1. Subtract H from each of the x-coordinates of the points on th graph of f. This results in a horizontal shift to the left if $H > 0$ or right if $H < 0$
2. Divide the x-coordinates of the points on the graph obtained in Step 1 by B. This results in a horizontal scaling, but may also include a reflection about the y-axis if $B < 0$

3. Multiply the y-coordinates of the points on the graph obtained in Step 2 by A. This results in a vertica scaling, but may also include a reflection about the x-axis if $A < 0$

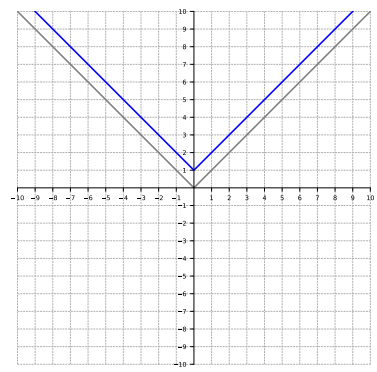
4. Add K to each of the y-coordinates of the points on the graph obtained in Step 3. This results in a vertical shift up if $K > 0$ or down if $K < 0$

1.7.1 Problems

1.

$$(2, 0)$$

19.



64.

$$g(x) = 2f(-x-3) - 1 = 2\sqrt[3]{-x-3} - 1$$

Chapter 2

Linear and Quadratic Functions

2.1 Linear Functions

Definition 2.1.1 (Slope)**.** The **slope** m of the line containing the points $P(x_0, y_0)$ and $Q(x_1, y_1)$ is:

$$m = \frac{y_1 - y_0}{x_1 - x_0} = \frac{\Delta y}{\Delta x}$$

provided $x_1 \neq x_0$.

Definition 2.1.2 (Point-slope form)**.** The **point-slope form** of the line with slope m containing the point (x_0, y_0) is the equation $y - y_0 = m(x - x_0)$.

Definition 2.1.3 (Slope-intercept form)**.** The **slope-intercept form** of the line with slope m and y-intercept $(0, b)$ is the equation $y = mx + b$.

Definition 2.1.4 (Linear function)**.** A **linear function** is a function of the form

$$f(x) = mx + b$$

where m and b are real numbers with $m \neq 0$. The domain of a linear function is $(-\infty, \infty)$.

Definition 2.1.5 (Constant function)**.** A **constant function** is a function of the form

$$f(x) = b$$

, where b is a real number. The domain of a constant function is $(-\infty, \infty)$.

Definition 2.1.6 (Average rate of change)**.** Let f be a function defined on the interval $[a, b]$. The **average rate of change** of f over $[a, b]$ is defined as:

$$\frac{\Delta f}{\Delta x} = \frac{f(b) - f(a)}{b - a}$$

. The line that connects $(a, f(a))$ with $(b, f(b))$ is called the **secant line**.

2.1.1 Exercises

1. $m=3$, $P(3,-1)$; point-slope: $y+1=3(x-3)$, slope-intercept: $y=3x-10$

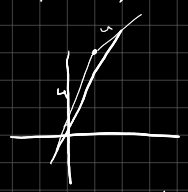
9. $m=-5$, $P(-\sqrt{3}, 2\sqrt{3})$; point-slope: $y-2\sqrt{3}=-5(x-\sqrt{3})$, slope-intercept: $y=-5x+7\sqrt{3}$

11. $P(0,0)$, $Q(-3,5)$; $m=\frac{\Delta y}{\Delta x}=\frac{5-0}{-3-0}=-\frac{5}{3}$ slope-intercept: $y=-5/3\,x$

19. $P(\sqrt{2},-\sqrt{2})$, $Q(-\sqrt{2},\sqrt{2})$; $m=\frac{\sqrt{2}-(-\sqrt{2})}{-\sqrt{2}-\sqrt{2}}=\frac{2\sqrt{2}}{-2\sqrt{2}}=-1$, slope-intercept: $y=-x+2\sqrt{2}$

21. $f(x)=2x-1$, $m=2$, y-intercept: -1, x-intercept: $1/2$

27. $y=2x+1$, $P(-1,3)$;



$$d=\sqrt{(x_1-x_0)^2+(y_1-y_0)^2}=\sqrt{(x_1+1)^2+(y_1-3)^2}$$
$$=\sqrt{(x+1)^2+(2x+1-3)^2}=\sqrt{x^2+2x+1+4x^2-8x+4}$$
$$=\sqrt{5x^2-6x+5}$$
$$16=5x^2-6x+5$$
$$5x^2-6x-11=0,\quad x_1=11/5,\quad x_2=-1,\quad y_1=\frac{27}{5},\quad y_2=-1$$

44. $f(x)=x^3$, $[-1,2]$, $\frac{\Delta y}{\Delta x}=\frac{8-(-1)}{2-(-1)}=3$

50. $f(x)=x^3$, $[x,x+h]$, $\frac{\Delta y}{\Delta x}=\frac{(x+h)^3-x^3}{x+h-x}=\frac{x^3+3x^2h+3xh^2+h^3-x^3}{h}=\frac{3x^2h+3xh^2+h^3}{h}=3x^2+3xh+h^2$

60. $y=-6x+5$, $P(3,2)$; $y_2=-6x+20$

67. $y=\frac{2}{3}x-7$, $P(6,0)$; $y_2=-\frac{3}{2}x+9$

2.2 Absolute Value Functions

Definition 2.2.1 (Absolute Value)**.** The **absolute value** of a real numberf x, denoted by $|x|$, is given by

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Definition 2.2.2 (Properties of Absolut Value)**.** Let a, b and x be real numbers and let n be an integer. Then

- **Product Rule**: $|ab| = |a||b|$
- **Power Rule**: $|a^n| = |a|^n$ whenever a^n is defined
- **Quotient Rule**: $|\frac{a}{b}| = \frac{|a|}{|b|}$, provided $b \neq 0$

Equality Properties:

- $|x| = 0$ if and only if $x = 0$.
- For $c > 0$, $|x| = c$ if and only if $x = c$ or $-x = c$.
- For $c < 0$, $|x| = c$ has no solution.

2.2.1 Exercises

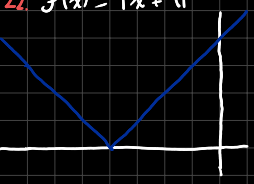
4. $4-|x|=3$, $x<0$: $4+x=3$, $x=-1$
$x\geq 0$: $4-x=3$, $x=1$

10. $|2x-1|=x+1$, $2x-1<0$, $-2x+1=x+1 \rightarrow 3x=0 \rightarrow x=0$
$2x-1\geq 0$, $2x-1=x+1 \rightarrow x=2$

15. $|x^2-1|=3$, $x^2-1<0$, $-x^2+1=3$, $\rightarrow x^2=-2$ $x\notin\mathbb{R}$
$x^2-1\geq 0$, $x^2-1=3 \rightarrow x^2=2 \rightarrow x_1=\sqrt{2}$, $x_2=-\sqrt{2}$

16. $|3x-2|=|2x+7|$ $|a|=|b| \rightarrow$ $a<0$ $a=-|b|=-b$; $|3x-2|=|2x+7|$, $3x-2\geq 0 \rightarrow 3x-2=2x+7 \rightarrow x=9$
$a>0$ $a=|b|=b$ $3x-2<0 \rightarrow 3x-2=-2x-7 \rightarrow x=1$

22. $f(x)=|x+4|$



x-intercepts: $(-4,0)$
y-intercepts: $(0,4)$

D: $(-\infty,\infty)$
R: $[0,\infty)$

Increasing on: $[-4,\infty)$
Decreasing on: $(-\infty,0]$

Relative and absolute minimum: $(-4,0)$
No relative or absolute maximum

2.3 Quadratic Functions

Definition 2.3.1 (Quadratic Function)**.** A **quadratic function** is a function of the form

$$f(x) = ax^2 + bx + c$$

, where a, b and c are real numbers with $a \neq 0$. The domain of a quadratic function is $(-\infty, \infty)$

Definition 2.3.2 (Standard and General Form of Quadratic Functions)**.** Suppose f is a quadratic function.

- The **general form** of the quadratic function f is $f(x) = ax^2 + bx + c$, where a, b and c are real numbers with $a \neq 0$
- The **standard form** of the quadratic function f is $f(x) = a(x-h)^2 + k$, where a, h and k are real numbers with $a \neq 0$

Definition 2.3.3 (Vertex Formula for Quadratics in Standard Form)**.** For the qudratic function $f(x) = a(x-h)^2 + k$, where a, h and k are real numbers with $a \neq 0$, the vertex of the graph $y = f(x)$ is (h, k)

Definition 2.3.4 (Vertex Formulas for Quadratic Functions)**.** Suppose a, b, c, h and k are real numbers with $a \neq 0$.

- If $f(x) = a(x-h)^2 + k$, the vertex of the graph of $y = f(x)$ is the point (h, k)
- If $f(x) = ax^2 + bx + c$, the vertex of the graph $y = f(x)$ is the point $(-\frac{b}{2a}, f(-\frac{b}{2a}))$

Definition 2.3.5 (The Quadratic Formula)**.** If a, b and c are real numbers with $a \neq 0$, then the solutions to $ax^2 + bx + c = 0$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

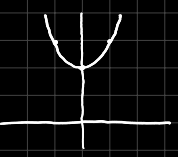
Definition 2.3.6 (Discriminant)**.** If a, b and c are real numbers with $a \neq 0$, then the **discriminant** of the quadratic equation $ax^2+bx+c = 0$ is the quantity $b^2 - 4ac$.

Definition 2.3.7 (Discriminant Trichotomy)**.** Let a, b and c are real numbers with $a \neq 0$.

- If $b^2 - 4ac < 0$, the equation $ax^2 + bx + c = 0$ has no real solutions.
- If $b^2 - 4ac = 0$, the equation $ax^2 + bx + c = 0$ has exactly one real solutions.
- If $b^2 - 4ac > 0$, the equation $ax^2 + bx + c = 0$ has exactly two real solutions.

2.3.1 Exercises

1. $f(x) = x^2 + 2$

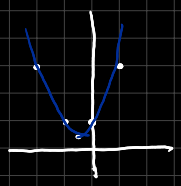


x-intercepts: None
y-intercepts: $(0,2)$

$f(x) = x^2 + 2 = 1(x+0)^2 + 2$

D: $(-\infty, \infty)$ Increasing: $[0, \infty)$ Vertex: $(0,2)$ Absolute Minimum
R: $[2, \infty)$ Decreasing: $(-\infty, 0]$ Symmetry: y-axis

7. $f(x) = x^2 + x + 1$



x-intercept: None
y-intercept: $(0,1)$

$f(x) = x^2 + x + 1 = 1(x + 1/2)^2 + 0$

D: $(-\infty, \infty)$ Increasing: $[-1/2, \infty)$ Vertex: $(-1/2, 3/4)$ Absolute Minimum
R: $[3/4, \infty)$ Decreasing: $(-\infty, -1/2)$ Symmetry: $x = -1/2$

15. $C(x) = 0.1x + 25$, $p(x) = 10 - 0.01x$, $P(x) = R(x) - C(x) = xp(x) - C(x) = x(10 - 0.01x) - (0.1x + 25) = 10x - 0.01x^2 - 0.1x - 25$
$= -0.01x^2 + 9.9x - 25$

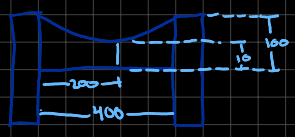
Vertex: $\left(\frac{9.9}{0.01}, P\left(\frac{9.9}{0.01}\right)\right) = (495, 2425.25)$, max profit with 495 cookies

21.

$\rightarrow A = ab$, $2a + 2b = 14 \rightarrow a = \frac{14 - 2b}{2} = 7 - b$, $A = (7-b)b = 7b - b^2$, $A' = 7 - 2b$, $7 - 2b = 0$, $b = 7/2$

$A(7/2) = 7(7/2) - (7/2)^2 = 49/4 = 12.25$

26.



$f(x) = ax^2 + bx + c$

$f(200) = 10$ $c = 100$
$f(0) = 100$ $(200)^2 a + 200b + c = 10$ $\rightarrow$ $a = \frac{9}{4000}$
$f(400) = 100$ $(400)^2 a + 400b + c = 100$ $b = -9/10$
$c = 100$

$f(x) = \frac{9}{4000}x^2 - \frac{9}{10}x + 100$, $f(50) = 66.625$

31. $x^2 - 10y^2 = 0$, $x^2 = 10y^2$, $x = \pm\sqrt{10}y$

35. $y^2 - 4y = x^2 - 4$, $y^2 - 4y - (x^2 - 4) = 0$ $y = \frac{+4 \pm \sqrt{16 + 4(x^2 - 4)}}{2} = \frac{4 \pm \sqrt{4x^2}}{2} = 2 \pm x$

2.4 Inequalities with Absolute Value and Quadratic Functions

Definition 2.4.1 (Graphical Interpretation of Equations and Inequalities)**.** Suppose f and g are functions.

- The solutions to $f(x) = g(x)$ are the x values where the graphs of $y = f(x)$ and $y = g(x)$ *intersect*.
- The solutions to $f(x) < g(x)$ are the x values where the graphs of $y = f(x)$ is *below* the graph of $y = g(x)$.
- The solutions to $f(x) \geq g(x)$ are the x values where the graphs of $y = f(x)$ is *above* the graph of $y = g(x)$.

Definition 2.4.2 (Inequalities Involving the Absolute Value)**.** Let c be a real number.

- For $c > 0$, $|x| < c$ is equivalent to $-c < x < c$.
- For $c > 0$, $|x| \leq c$ is equivalent to $-c \leq x \leq c$.
- For $c \leq 0$, $|x| < c$ has no solution, and for $c < 0$, $|x| \leq c$ has no solution.
- For $c \geq 0$, $|x| > c$ is equivalent to $x < -c \cup x > c$.
- For $c \geq 0$, $|x| \geq c$ is equivalent to $x \leq -c \cup x \geq c$.
- For $c < 0$, $|x| > c$ and $|x| \geq c$ are true for all real numbers.

2.4.1 Exercises

1. $|3x-5| \le 4$, $3x-5 \ge 0 \to 3x-5 \le 4$, $3x \le 9$, $x \le 3$ $\quad \frac{1}{3} \le x \le 3$, $[\frac{1}{3}, 3]$

$3x-5 < 0 \to -3x+5 \le 4$, $-3x \le -1$, $x \ge 1/3$

31. $x|x+5| \ge -6$, $x+5 \ge 0 \to x(x+5) \ge -6$, $x^2+5x+6 \ge 0$, $x \le -3$, $x \ge -2$ $\quad x \ge -5$

$x+5 < 0 \to x(-x-5) \ge -6$, $-x^2-5x+6 \ge 0$, $-6 \le x \le 1$ $\quad x < -5$

$\} \to [-5,-3] \cup [-2,\infty)$ $\to [-6,-3] \cup [-2,\infty)$

$[-6,-5)$

36. $h(t) = -5t^2+100t$, $0 \le t \le 20$

$-5t^2+100t = 250$, $t = \frac{-100 \pm \sqrt{100^2-4(-5)(-250)}}{2(-5)}$, $t = 10 \pm 5\sqrt{2}$ $\quad [10-5\sqrt{2}, 10+5\sqrt{2}]$

43. $S(x) = 6x^2$, $x > 0$ $\quad |6x^2-42| \le 3$, $6x^2-42 \ge 0 \to 6x^2-42 \le 3 \to 6x^2-45 \le 0$, $(\frac{-\sqrt{30}}{2}, \frac{\sqrt{30}}{2})$

$6x^2-42 < 0 \to -6x^2+42 \le 3 \to -6x^2+39 \le 0$ $\quad \cancel{(\frac{-\sqrt{26}}{2}, \frac{\sqrt{26}}{2})}$ $\to (-\infty, -\frac{\sqrt{26}}{2}) \cup (\frac{\sqrt{26}}{2}, \infty)$ $\} \quad [\frac{\sqrt{26}}{2}, \frac{\sqrt{30}}{2}]$

Chapter 3

Polynomial Function

3.1 Graphs of Polynomials

Definition 3.1.1 (Polynomial Function)**.** A **polynomial function** is a function of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdot + a_2x^2 + a_1x + a_0$$

where $a_0, a_1, \cdot, a_n$ are real numbers and $n \geq 1$ is a natural number. The domain of a polynomial function is $(-\infty, \infty)$.

Definition 3.1.2 (Properties of polynomial functions)**.** Suppose f is a polynomial function.

- Given $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdot + a_2x^2 + a_1x + a_0$ with $a_n \neq 0$, we say
  - The natural number n is called the **degree** of the polynomial f.
  - The term $a_n x^n$ is called the **leading term** of the polynomial f.
  - The real number a_n is called the **leading coefficient** of the polynomial f.
  - The reaul number a_0 is called the **constant term** of the polynomial f.
- If $f(x) = a_0$, and $a_0 \neq 0$, we say f has degree 0.
- If $f(x) = 0$, we say f has no degree.

Definition 3.1.3 (End Behavior of Functions $f(x) = ax^n$, n is even)**.** Suppose $f(x) = ax^n$, where $a \neq 0$ is a real number and n is an even natural number. The end behavior of th graph $y = f(x)$ matches one of the following:

- for $a > 0$, as $x \to -\infty$, $f(x) \to \infty$ and as $x \to \infty$, $f(x) \to \infty$
- for $a < 0$, as $x \to -\infty$, $f(x) \to -\infty$ and as $x \to \infty$, $f(x) \to -\infty$

Definition 3.1.4 (End Behavior of Functions $f(x) = ax^n$, n is odd)**.** Suppose $f(x) = ax^n$, where $a \neq 0$ is a real number and $\geq 3n$ is an odd natural number. The end behavior of th graph $y = f(x)$ matches one of the following:

- for $a > 0$, as $x \to -\infty$, $f(x) \to -\infty$ and as $x \to \infty$, $f(x) \to \infty$
- for $a < 0$, as $x \to -\infty$, $f(x) \to \infty$ and as $x \to \infty$, $f(x) \to -\infty$

Definition 3.1.5 (The Intermediate Value Theorem)**.** Suppose f is a continuous function on an interval containing $x = a$ and $x = b$ with $a < b$. If $f(a)$ and $f(b)$ have different signs, then f has at least one zero between $x = a$ and $x = b$; that is, for at least one real number c such that $a < c < b$, we have $f(c) = 0$.

Definition 3.1.6 (End Behavior for Polynomial Functions)**.** The end behavior of a polynomial $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdot + a_1 x + a_0$ with $a_n \neq 0$ matches the end behavior of $y = a_n x^n$.

Definition 3.1.7 (Multiplicity)**.** Suppose f is a polynomial functiuon and m is a natural number. If $(x - c)^m$ is a factor of $f(x)$ but $(x - c)^{m+1}$ is not, then we say $x = c$ is a zero of **multiplicity** m.

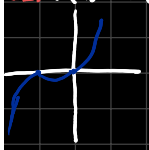
Definition 3.1.8 (Role of multiplicity)**.** Suppose f is a polynomial function and $x = c$ is a zero of multiplicity m.

- If m is even, the graph of $y = f(x)$ touches and rebounds from the x-axis at $(c, 0)$.
- If m is odd, the graph of $y = f(x)$ crosses through the x-axis at $(c, 0)$.

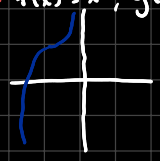
3.1.1 Problems

1. $f(x) = 4 - x - 3x^2$, degree: 2, leading term: $-3x^2$, constant term: 4, end behavior: $x \to -\infty, f(x) \to -\infty$; $x \to \infty, f(x) \to -\infty$

11. $a(x) = x(x+2)^2$, z: $x = 0$, $x = -2$



21. $f(x) = x^3$, $g(x) = (x+2)^3 + 1$; D: $(-\infty, \infty)$ R: $(-\infty, \infty)$

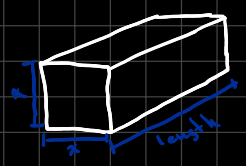


$(0,0) \to (-2,1)$
$(1,1) \to (-1,2)$
$(-1,-1) \to (-3,0)$

27. $f(x) = x^3 - 9x + 5$; $[-4,-3]$, $[0,1]$, $[2,3]$

$\left.\begin{matrix} f(-4) = -23 \\ f(-3) = 5 \end{matrix}\right\}$ sign change $\quad \left.\begin{matrix} f(0) = 5 \\ f(1) = -3 \end{matrix}\right\}$ sign change $\quad \left.\begin{matrix} f(2) = -5 \\ f(3) = 5 \end{matrix}\right\}$ sign change

33. $\underbrace{\text{Length} + \text{Girth} \leq 130}_{\text{Parcel Post}}$ $\quad \underbrace{\text{Length} + \text{Girth} \leq 108}_{\text{Other}}$



$G(x) = 4x$

a) $L + G = 130 \to L = 130 - G(x) = 130 - 4x$
$V = (L)(x)(x) = Lx^2 = (130 - 4x)x^2 = 130x^2 - 4x^3$

b) $V_{max} = (21.66\ldots, 20342.59)$

c) $L + G = 108$, $L = 108 - G(x) = 108 - 4x$; $V(x) = L(x)(x)(x) = (108 - 4x)x^2 = 108x^2 - 4x^3$

$V_{max} = (18, 11664)$

3.2 The Factor Theorem and the Remainder Theorem

Definition 3.2.1 (Polynomial Division)**.** Suppose $d(x)$ and $p(x)$ are nonzero polynomials where the degree of p is greater than or equal to the degree of d. There exist two unique polynomials, $q(x)$ and $r(x)$ such that $p(x) = d(x)q(x) + r(x)$, where either $r(x) = 0$ or the degree of r is strictly less than the degree of d.

Definition 3.2.2 (The Remainder Theorem)**.** Suppose p is a polynomial of degree at least 1 and c is a real number. When $p(x)$ is divided by $x - c$ the remainder is $p(c)$.

Definition 3.2.3 (The Factor Theorem)**.** Suppose p is a nonzero polynomial. The real number c is a zero of p if and only if $(x - c)$ is a factor of $p(x)$.

Definition 3.2.4 (Number of zeros)**.** Suppose f is a polynomial of degree $n \geq 1$. Then f has at most n real zeros, counting multiplicities.

Definition 3.2.5 (Connections Between Zeros, Factors and Graphs of Polynomial Functions)**.** Suppose p is a polynomial function of degree $n \geq 1$. The following statements are equivalent:

- The real number c is a zero of p.
- $p(c) = 0$.
- $x = c$ is a solution to the polynomial equation $p(x) = 0$.
- $(x - c)$ is a factor of $p(x)$.
- The point $(c, 0)$ is an x-intercept of the graph of $y = p(x)$.

3.2.1 Exercises

1. $(4x^2+3x-1)/(x-3) = (4x+15)+44$

$$
\begin{array}{r|l}
 & 4x+15 \\
\hline
x-3 & 4x^2+3x-1 \\
 & -4x^2+12x \\
\hline
 & +15x-1 \\
 & -15x \\
\hline
 & -1 \\
 & +45 \\
\hline
 & +44
\end{array}
$$

7. $(3x^2-2x+1)/(x-1) = (3x+1)+2/(x-1)$

$$
\begin{array}{r|rrr}
1 & 3 & -2 & 1 \\
 & & 3 & 1 \\
\hline
 & 3 & 1 & 2
\end{array}
$$

9. $(-2x^2-4x+3)/(x+1) = (-2x-2)+5/(x+1)$

$$
\begin{array}{r|rrr}
-1 & -2 & -4 & 3 \\
 & & 2 & 2 \\
\hline
 & -2 & -2 & 5
\end{array}
$$

11. $p(x)=2x^2-x+1$, $c=4$; $p(4)=2(4)^2-4+1=32-4+1=29$; $2x^2-x+1=(x-4)(2x+7)+29$

31. $x^3-6x^2+11x-6$, $c=1$; $x^3-6x^2+11x-6=(x-1)(x^2-5x+6)=(x-1)(x-2)(x-3)$

35. x^3+2x^2-3x-6, $c=-2$; $\rightarrow (x+2)(x^2-3)=(x+2)(x+\sqrt{3})(x-\sqrt{3})$

41. $p(2)=0$, $p(-2)=0$, $p(1)=0$, $p(-1)=0$

$p(x)=117(x-2)(x+2)(x-1)(x+1)$

3.3 Real Zeros of Polynomials

Theorem 3.3.1 (Cauchy's Bound)**.** *Suppose* $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdot + a_2x^2 + a_1x + a_0$ *is a polynomial of degree* n *with* $n \geq 1$*. Let* M *be the largest of the numbers:* $\frac{|a_0|}{a_n}, \frac{|a_1|}{a_n}, \cdot, \frac{|a_{n-1}|}{a_n}$*. Then all the real zeros of* f *lie in the interval* $[-(M+1), M+1]$

Theorem 3.3.2 (Rational Zeros Theorem)**.** *Suppose* $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdot + a_2x^2 + a_1x + a_0$ *is a polynomial of degree* n *with* $n \geq 1$*, and* $a_0, a_1, \cdot, a_n$ *are integers. If* r *is a rational zero of* f*, then* r *is of the form* $\pm\frac{p}{q}$*, where* p *is a factor of the constant term* a_0*, and* q *is a factor of the leading coefficient* a_n*.*

Theorem 3.3.3 (Descartes' Rule of Signs)**.** *Suppose* $f(x)$ *is the formula for a polynomial function written with descending powers of* x*.*

- *If* P *denotes the number of variations of sign in the formula for* $f(x)$*, then the number of positive real zeros (counting multiplicity) is one of the numbers* $\{P, P-2, P-4, \cdot\}$
- *If* N *denotes the number of variations of sign in the formula for* $f(-x)$*, then the number of negative real zeros (counting multiplicity) is one of the numbers* $\{N, N-2, N-4, \cdot\}$

Theorem 3.3.4 (Upper and Lower Bounds)**.** *Suppose* f *is a polynomial of degree* $n \geq 1$*.*

- *If* $c > 0$ *is synthetically divided into* f *and all of the numbers in the final line of the division tableaut have the same signs, then* c *is an upper bound for the real zeros of* f*. That is, there are no real zeros greater than* c*.*
- *If* $c < 0$ *is synthetically divided into* f *and all of the numbers in the final line of the division tableaut have the same signs, then* c *is an upper bound for the real zeros of* f*. That is, there are no real zeros less than* c*.*
- *If* $c = 0$ *occurs in the final line of the division tableau in either of the above cases, it can be treated as positive or negative as needed.*

3.3.1 Exercises

1. $f(x) = x^3 - 2x^2 - 5x + 6$, $M = \max(2, 5, 6) = 6$ $Z: [-5, 7]$

Possible rational zeros: $[\pm 6, \pm 3, \pm 2, \pm 1]$

Possible number of positive real zeros: $[2]$ Possible number of negative real zero: $[1]$

11. $f(x) = x^3 - 2x^2 - 5x + 6$;

$(x-3)(x-1)(x+2)$

31. $f(x) = x^5 - 60x^3 - 80x^2 + 960x + 2304 \rightarrow x = 6$ (mult. 2), $x = -4$ (mult. 3)

35. $9x^3 = 5x^2 + x$, $9x^3 - 5x^2 - x = x(9x^2 - 5x - 1) = 0$, $x = 0$ (mult. 1), $x = \frac{5 \pm \sqrt{61}}{18}$

45. $-2x^3 + 19x^2 - 49x + 20 > 0$ $x = 5$, $x = 4$, $x = 0.5$, $(-\infty, 0.5) \cup (4, 5)$

3.4 Complex Zeros and the Fundamental Theorem of Algebra

Definition 3.4.1 (Imaginary Unit)**.** The imaginary unit i satisfies the two following properties

- $i^2 = -1$
- If c is a real number with $c \geq 0$ then $\sqrt{-c} = i\sqrt{c}$

Definition 3.4.2 (Complex number)**.** A complex number is a number of the form $a + bi$, where a and b are real numbers and i is the imaginary unit.

Theorem 3.4.3 (Properties of the Complex Conjugate)**.** *Let z and w be complex numbers.*

- $\overline{\overline{z}} = z$
- $\overline{z} + \overline{w} = \overline{z + w}$
- $\overline{zw} = \overline{z}\,\overline{w}$
- $(\overline{z})^n = \overline{z^n}$, *for any natural number n*
- *z is a real number if and only if $\overline{z} = z$*

Theorem 3.4.4 (The Fundamental Theorem of Algebra)**.** *Suppose f is a polynomial function with complex number coefficients of degree $n \geq 1$, then f has at least one complex zero.*

Theorem 3.4.5 (Complex Factorization Theorem)**.** *Suppose f is a polynomial function with complex number coefficients. If the degree of f is n and $n \geq 1$, then f has exactly n complex zeros, counting multiplicity. If $z_1, z_2, \cdot, z_n$ are the distinct zeros of f, with multiplicities $m_1, m_2, \cdot, m_k$, respectively, then $f(x) = a(x - z_1)^{m_1}(x - z_2)^{m_2}(x - z_3)^{m_3} \cdot (x - z_k)^{m_k}$*

Theorem 3.4.6 (Conjugate Pairs Theorem)**.** *If f is a polynomial function with real number coefficients and z is a zero of f, then so is $\overline{z}$*

Theorem 3.4.7 (Real Factorization Theorem)**.** *Suppose f is a polynomial function with real number coefficients. Then $f(x)$ can be factored into a product of linear factors corresponding to the real zeros of f and irreducible quadratic factors which give the nonreal zeros of f.*

3.4.1 Exercises

1. $z = 2+3i$, $w = 4i$ • $z+w = 2+7i$ • $1/z = \frac{1}{2+3i} = \frac{2-3i}{13}$ • $\bar{z} = 2-3i$ • $zw = (2+3i)(4i) = -12+8i$ • $z/w = \frac{-12+8i}{-16}$

• $z\bar{z} = (2+3i)(2-3i) = 13$ • $z^2 = -5+12i$ • $w/z = \frac{4i}{2+3i} = \frac{12+8i}{13}$ • $(\bar{z})^2 = -5-12i$

11. $\sqrt{-49} = \sqrt{-(7^2)} = 7\sqrt{-1} = 7i$

19. $i^5 = i^2 i^2 i = (-1)(-1)i = i$

27. $f(x) = x^2 - 4x + 13 = (x-(2+3i))(x-(2-3i))$

49. $f(x) = 42(x+1)(x-1)(x+i)(x-i)$

Chapter 4

Rational Functions

4.1 Introduction to Rational Functions

Definition 4.1.1 (Rational function)**.** A **rational function** is a function which is the ratio of polynomial functions. Said differently, r is a rational function if it is of thje form

$$r(x) = \frac{p(x)}{q(x)}$$

where p and q are polynomial functions.

Definition 4.1.2 (Asymptote)**.** Asymptotes are lines that a curve approaches initely close as the function heads towards a point.

- The line $x = c$ is called a **vertical asymptote** of the graph of a function $y = f(x)$ if as $x \to c^-$ or as $x \to c^+$, either $f(x) \to \infty$ or $f(x) \to -\infty$
- The line $y = c$ is called a **horizontal asymptote** of the graph of a function $y = f(x)$ if as $x \to -\infty$ or as $x \to \infty$, $f(x) \to c$

Theorem 4.1.3 (Location of Vertical Asymptotes and Holes)**.** *Suppose r is a rational function which can be written as $r(x) = \frac{p(x)}{q(x)}$ where p and q have no common zeros. Let c be a real number which is not in the domain of r.*

- *If $q(c) \neq 0$, then the graph of $y = r(x)$ has a hole at $(q, \frac{p(c)}{q(c)})$.*
- *If $q(c) = 0$, then the line $x = c$ is a vertical asymptote of the graph of $y = r(x)$*

Theorem 4.1.4 (Location of Horizontal Asymptotes)**.** *Suppose r is a rational function and $r(x) = \frac{p(x)}{q(x)}$, where p and q are polynomial functions with leading coefficients a and b, respectively.*

- *If the degree of $p(x)$ is the same degree of $q(x)$, then $y = \frac{a}{b}$ is the horizontal asymptote of the graph of $y = r(x)$.*

- *If the degree of $p(x)$ is less than the degree of $q(x)$, then $y = 0$ is the horizontal asymptote of the graph of $y = r(x)$*
- *If the degree of $p(x)$ is greater than the degree of $q(x)$, then the graph of $y = r(x)$ has no horizontal asymptotes.*

Definition 4.1.5 (Slant asymptote)**.** The line $y = mx+b$ where $m \neq 0$ is called a **slant asymptote** of the graph of a function $y = f(x)$ if as $x \to -\infty$ or as $x \to \infty$, $f(x) \to mx + b$

Theorem 4.1.6 (Determination of Slant Asymptotes)**.** *Suppose r is a rational function and $r(x) = \frac{p(x)}{q(x)}$, where the degree of p is exactly one more than the degree of q. Then the graph of $y = r(x)$ has the slant asymptote $y = L(x)$ where $L(x)$ is the quotient obtained by dividing $p(x)$ by $q(x)$.*

4.1.1 Exercises

1. $f(x) = \frac{x}{3x-6}$, $D: (-\infty, 3) \cup (3, \infty)$ Vertical Asymptote: $x = 3$ Holes: None

Horizontal Asymptote: $y = 1/3$

Slant Asymptote: None

19. $C(p) = \frac{1770p}{100-p}$, $0 \le p < 100$

a) $C(25) = \frac{1770(25)}{100-25} = 590$, $C(95) = 33630$

b) Impossible to remove 100%.

c) $40{,}000 = \frac{1770p}{100-p}$, $40{,}000(100-p) = 1770p$, $4000000 - 40{,}000p = 1770p$, $41770p = 4{,}000{,}000$

$p = \frac{4{,}000{,}000}{41770} \approx 95.76$

20. $P(t) = \frac{150t}{t+15}$, Horizontal Asymptote: $P = 150$; Population maxes out at 150

4.2 Graphs of Rational Functions

Procedure 4.2.1 (Steps for Consutructing a Sign Diagram for a Rational Function)**.** Suppose r is a rational function.

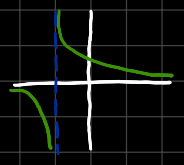
1. Place any values excluded from the domain r on the number line with distinguishable sign.
2. Find the zeros of r and place them on the number line with the number 0 above them.
3. Choose a test value in each of the intervals determined in the previous steps.
4. Determine the sign of $r(x)$ for each test value in the previous step, and write that sign above the corresponding interval.

Procedure 4.2.2 (Steps for Graphing Rational Functions)**.** Suppose that r is a rational function.

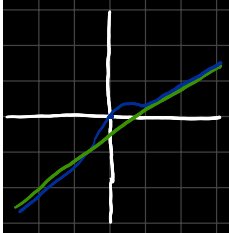
1. Find the domain of r.
2. Reduce $r(x)$ to lowest terms, if applicable.
3. Find the x- and y-intercepts of the graph of $y = r(x)$, if they exist.
4. Determine the location of any verical asymptotes or holes in the graph, if they exist. Analyze the behavior of r on either side of the vertical asymptotes, if applicable.
5. Analyze the end behavior of r. Find the horizontal or slant asymptote, if one exists.
6. Use a sign diagram and plot additional points, as needed, to sketch the graph of $y = r(x)$

4.2.1 Exercises

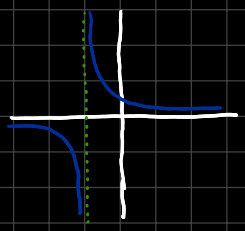
1. $f(x) = \frac{4}{x+2}$, vertical asymptotes: $x=-2$



15. $f(x) = \frac{x^3-2x^2+3x}{2x^2+2} = \frac{x(x^2-2x+3)}{2x^2+2}$ slant asymptote: $y = \frac{x}{2} - 1$



17. $f(x) = \frac{1}{x-2}$



4.3 Rational Inequalities and Applications

4.3.1 Variation

Definition 4.3.1 (Variations)**.** Suppose x, y and z are variable quantities. We say

- y **varies directly with** (or is **directly proportional to**) x if there is a constant k such that $y = kx$
- y **varies inversely with** (or is **inversely proportional to**) x if there is a constant k such that $y = \frac{k}{x}$.
- z **varies jointly with** (or is **jointly proportional to**) x and y if there is a constant k such that $z = kxy$.

The constant k in the above definitions is called the **constant of proportionality**

4.3.2 Exercises

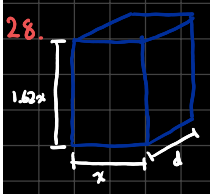
1. $\frac{x}{5x+4}=3$; $x=3(5x+4)=15x+12$, $14x=-12$, $x=\frac{-12}{14}=\frac{-6}{7}$

7. $\frac{1}{x+2}\geq 0$; $(-2,\infty)$

13. $\frac{x^3+2x^2+x}{x^2-x-2}\geq 0$; $\underset{-1}{\cdot}\ (+)\ \underset{0}{\cdot}\ (-)\ \underset{2}{\cdot}\ (+)$, $(-1,0]\cup(2,\infty)$

21. $\frac{3}{6}+\frac{1}{6}=\frac{3}{c}$, $\frac{3}{c}=4/6=\frac{2}{3}$, $3=\frac{2}{3}c$, $c=\frac{9}{2}=4.5$

28.



$140=(1.62x)(x)(d)$

$d=\frac{86.42}{x^2}$

$S(x)=2(1.62x)(x)+2\left(\frac{86.42}{x^2}\right)(1.62x)+2(x)\left(\frac{86.42}{x^2}\right)$

$=3.24x^2+\frac{280x}{x^2}+\frac{172.84x}{x^2}=3.24x^2+\frac{452.84x}{x^2}$

$=\frac{3.24x^3+452.84}{x}$

$S_{min}: x\approx 4.12 \rightarrow h\approx 6.67 \quad d\approx 5.09$

31.

$7.35=\pi r^2 h$ $\rightarrow h=\frac{7.35}{\pi r^2}$

$S=2\pi rh+2\pi r^2$

$S(r)=2\pi r\left(\frac{7.35}{\pi r^2}\right)+2\pi r^2$

$=\frac{14.7}{r}+2\pi r^2$

$S_{min}: r\approx 1.05$, $h\approx 2.12$

Chapter 5

Further Topics in Functions

5.1 Function Composition

Definition 5.1.1 (Composite function)**.** Suppose f and g are two functions. The **composite** if g with f, denoted $f \circ g$ is defined by the formula $(g \circ f)(x) = g(f(x))$, provided x is an element of the domain of f and $f(x)$ is an element of the domain of g

Theorem 5.1.2 (Properties of Function Composition)**.** *Suppose f, g, and h are functions.*

- $h \circ (g \circ f) = (h \circ g) \circ f$*, provided the composite functions are defined.*
- *If I is defined as $I(x) = x$ for all real numbers x, then $I \circ f = f \circ I = f$*

5.1.1 Exercises

1. $f(x)=x^2$, $g(x)=2x+1$; • $(g\circ f)(0)=2(0^2)+1=1$ • $(f\circ g)(-1)=(2(-1)+1)^2=1$ • $(f\circ f)(2)=(2^2)^2=16$ • $(g\circ f)(-3)=2(-3)^2+1$
$=18+1=19$ • $(f\circ g)(1/2)=(2(1/2)+1)^2=4$ • $(f\circ f)(-2)=((-2)^2)^2=16$

13. $f(x)=2x+3$, $g(x)=x^2-9$; • $(g\circ f)(x)=(2x+3)^2-9=4x^2+12x$ D: $(-\infty,\infty)$ • $(f\circ g)(x)=2(x^2-9)+3=2x^2-15$ D: $(-\infty,\infty)$
• $(f\circ f)(x)=2(2x+3)+3=4x+9$ D: $(-\infty,\infty)$

25. $f(x)=-2x$ $g(x)=\sqrt{x}$ $h(x)=|x|$. $(h\circ g\circ f)(x)$, $(g\circ f)(x)=\sqrt{-2x}$ D: $(-\infty,0]$
$(h\circ(g\circ f))(x)=|\sqrt{-2x}|=\sqrt{-2x}$ D: $(-\infty,0]$

31. $p(x)=(2x+3)^3$; $g(x)=x^3$, $f(x)=2x+3$; $(g\circ f)(x)=(2x+3)^3$

63. $x(t)=-4t^2+20t+92$, $0\le t\le 4$
a) $R(x)=2x$
b) $(R\circ x)(t)=2(-4t^2+20t+92)=-8t^2+40t+184$, Revenue per hour
c) $(R\circ x)(2)=-8(2)^2+40(2)+184=-8(4)+80+184=-32+264=232$

5.2 Inverse Functions

Definition 5.2.1 (Inverse functions)**.** Suppose f and g are two functions such that

1. $(g \circ f)(x) = x$ for all x in the domain of f **and**
2. $(f \circ g)(x) = x$ for all x in the domain of g

then f and g are **inverses** of each other and the functions f and g are said to be **invertible**.

Theorem 5.2.2 (Properties of Inverse Functions)**.** *Suppose f and g are inverse functions.*

- *The range of f is the domain of g and the domain of f is the range of g*
- *$f(a) = b$ if and only if $g(b) = a$*
- *(a, b) is on the graph of f if and only (b, a) is on the graph of g*

Theorem 5.2.3 (Uniqueness of Inverse Functions and Their Graphs)**.** *Suppose f is an invertible function.*

- *There is exactly one inverse function for f, denoted f^-1*
- *The graph of $y = f^-1(x)$ is the reflection of the graph $y = f(x)$ across the line $y = x$*

Definition 5.2.4 (One-to-One Functions)**.** A function f is said to be **one-to-one** if f matches different inputs to different outputs. Equivalently, f is one-to-one if and only if whenever $f(c) = f(d)$, then $c = d$

Theorem 5.2.5 (The Horizontal Line Test)**.** *A function f is one-to-one if and only if no horizontal line intersects the graph of f more than once.*

Theorem 5.2.6 (Equivalent Conditions for Invertibility)**.** *Suppose f is a function. The following statements are equivalent.*

- *f is invertible.*
- *f is one-to-one*
- *The graph of f passes the Horizontal Line Test*

Procedure 5.2.7 (Steps for finding the Inverse of a One-to-one Function)**.**

1. Write $y = f(x)$
2. Interchange x and y
3. Solve $x = f(y)$ for y to obtain $y = f^-1(x)$

5.2.1 Exercises

1. $f(x)=6x-2$ D: $(-\infty,\infty)$ R: $(-\infty,\infty)$; $f^{-1}(x)=\frac{x+2}{6}$ D: $(-\infty,\infty)$ R: $(-\infty,\infty)$

$x=f(y)=6y-2 \rightarrow y=\frac{x+2}{6}=f^{-1}(x)$

21. $f(x)=ax+b,\ a\neq 0$; $x=f(y)=ay+b,\ y=\frac{x-b}{a}=f^{-1}(x)$

25. $p(x)=450-15x,\ 0\leq x\leq 30$;

a) $x=p(y)=450-15y,\ y=\frac{x-450}{-15}=p^{-1}(x)$ D: $[0,450]$

b) $p^{-1}(105)=\frac{105-450}{-15}=\frac{-345}{-15}=+23$, The price is 105 when 23 are sold weekly

c) $P(x)=-15x^2+350x-2000,\ 0\leq x\leq 30$

$(P\circ p^{-1})(x)=-15\left(\frac{450-x}{15}\right)^2+350\left(\frac{450-x}{15}\right)-2000$, $(P\circ p^{-1})(x)_{max}: x=275 \rightarrow 41.66$

$p^{-1}(275)=11.66$, $p(11)=285$, $p(12)=270$ $(P\circ p^{-1})(285)=35$, $(P\circ p^{-1})(270)=\underline{40}$

5.3 Other Algebraic Functions

Definition 5.3.1 (Radicals)**.** Let x be a real number and n a natural number. If n is odd, the p**principal** n^{th} **root** of x, denoted by $\sqrt[n]{x}$ is the unique real number satisfying $(\sqrt[n]{x})^2 = x$. If n is even, $\sqrt[n]{x}$ is defined similarly provided $x \geq 0$ and $\sqrt[n]{x} \geq 0$. The **index** is the number n and the **radicand** is the number x. For $n = 2$, we write $\sqrt{x}$ instead of $\sqrt[2]{x}$.

Theorem 5.3.2 (Properties of Radicals)**.** *Let x and y be real numbers and m and n be natural numbers. If $\sqrt[n]{x}$, $\sqrt[n]{y}$ are real numbers, then*

- ***Product Rule**: $\sqrt[n]{xy} = \sqrt[n]{x}\sqrt[n]{y}$*
- ***Powers of Radicals**: $\sqrt[n]{x^m} = (\sqrt[n]{x})^m$*
- ***Quotient Rule**: $\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$*
- *If n is odd, $\sqrt[n]{x^n}$; if n is even, $\sqrt[n]{x^n} = |x|$*

Definition 5.3.3 (Exponential notation for radicals)**.** Let x be a real number, m an integer and n a natural number.

- $x^{\frac{1}{n}} = \sqrt[n]{x}$ and is defined whenever $\sqrt[n]{x}$ is defined.
- $x^{\frac{m}{n}} = (\sqrt[n]{x})^m$, whenever $(\sqrt[n]{x})^m$ is defined.

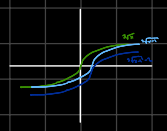
Procedure 5.3.4. Suppose f is an algebraic function.

- Place any values excluded from the domain of f on the number line with a distinguishable above them.
- Find the zeros of f and place them on the number line with the number 0 above them.
- Choose a test value in each of the intervals determined previously.
- Determine the sign of $f(x)$ for each test value in the previous step, and write that sign above the corresponding interval.

5.3.1 Exercises

1. $f(x)=\sqrt{1-x^2}$; $1-x^2=0$, $x=\pm 1$ • D: $[-1,1]$ •

11. $f(x)=\sqrt[3]{x}$, $g(x)=\sqrt[3]{x-1}-2$



17.

$$x+1=\sqrt{3x+7}$$
$$(x+1)^2=3x+7$$
$$x^2+2x+1=3x+7$$
$$x^2-x-6=0$$
$$(x-3)(x+2)=0$$
$$x=3$$

25. $\sqrt{2x+1}=3+\sqrt{4-x}$

$$2x+1=9+6\sqrt{4-x}+4-x$$
$$3x-12=6\sqrt{4-x}$$
$$x-4=2\sqrt{4-x}$$
$$x=4$$

40. $T=2\pi\sqrt{\frac{L}{g}}$, $\frac{1}{2}=2\pi\sqrt{L/g}$, $\frac{1}{4\pi}=\sqrt{L/g}$, $\frac{1}{16\pi^2}=\frac{L}{g}$, $L=\frac{g}{16\pi^2}=0.062m=62mm$

Chapter 6

Exponential and Logarithmic Functions

6.1 Introduction to Exponentail and Logarithmic Functions

Definition 6.1.1 (Exponential Function)**.** A function of the form $f(x) = b^x$ where b is a fixed real number, $b > 0$, $b \neq 1$ is called a **base** b **exponential function**.

Theorem 6.1.2 (Properties of Exponential Functions)**.** *Suppose* $f(x) = b^x$.

- *The domain of* f *is* $(-\infty, \infty)$ *and the range of* f *is* $(0, \infty)$.
- $(0, 1)$ *is on the graph of* f *and* $y = 0$ *is a horizontal asymptote to the graph of* f.
- f *is one-to-one, continuous and smooth.*
- *If* $b > 1$*:*
  - f *is always increasing*
  - *As* $x \to -\infty$, $f(x) \to 0^+$
  - *As* $x \to \infty$, $f(x) \to \infty$
- *If* $0 < b < 1$*:*
  - f *is always decreasing*
  - *As* $x \to -\infty$, $f(x) \to \infty$
  - *As* $x \to \infty$, $f(x) \to 0^+$

Definition 6.1.3 (Logarithmic Function)**.** The inverse of the exponential function $f(x) = b^x$ is called the **base** b **logarithmic function**, and is denoted by $f^{-1}(x) = \log_b(x)$. We read $\log_b(x)$ as log base b of x.

Definition 6.1.4. The **common logarithm** of a real number x is $\log_{10}(x)$ and is usually written $\log(x)$. The **natural logarithm** of a real number x is $\log_e(x)$ and is usually written $ln(x)$.

Theorem 6.1.5 (Properties of Logarithmic Functions)**.** *Suppose* $f(x) = \log_b(x)$

- *The domain of* f *is* $(0, \infty)$ *and the range of* f *is* $(-\infty, \infty)$.
- $(1, 0)$ *is on the graph of* f *and* $y = 0$ *is a vertical asymptote to the graph of* f.
- f *is one-to-one, continuous and smooth.*
- $b^a = c$ *if and only if* $\log_b(c) = a$. *That is,* $\log_b(c)$ *is the exponent you put on* b *to obtain* c.
- $\log_b(b^x) = x$ *for all* x *and* $b^{\log_b(x)} = x$ *for all* $x > 0$
- *If* $b > 1$*:*
  - f *is always increasing*
  - *As* $x \to 0^+$, $f(x) \to -\infty$
  - *As* $x \to \infty$, $f(x) \to \infty$
- *If* $0 < b < 1$*:*
  - f *is always decreasing*
  - *As* $x \to 0^+$, $f(x) \to \infty$
  - *As* $x \to \infty$, $f(x) \to -\infty$

6.1.1 Exercises

1. $2^3 = 8$, $\log_2(8) = 3$

4. $\left(\frac{1}{3}\right)^{-2} = 9$, $\log_{1/3}(9) = -2$

16. $\log_3(27) = \log_3(3^3) = 3\log_3(3) = 3$

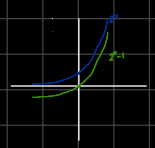
31. $\log_{36}(\sqrt[4]{36}) = \log_{36}(36^{1/4}) = \frac{1}{4}\log_{36}(36) = 1/4$

43. $f(x) = \ln(x^2+1)$, $x^2+1>0$; $D: (-\infty, \infty)$

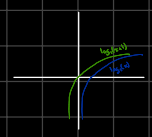
45. $f(x) = \ln(4x-20) = \ln(4(x-5)) = \ln 4 + \ln(x-5)$, $x-5>0$, $x>5$; $D: (5, \infty)$

53. $f(x) = \log_9(|x+3|-4)$, $|x+3|-4>0$, $|x+3|>4$, $x+3>4 \to x>1$; $D: (-\infty,-7) \cup (1,\infty)$
$-x-3>4 \to x<-7$

58. $f(x) = 2^x$, $g(x) = 2^x - 1$



64. $f(x) = \log_2(x)$, $g(x) = \log_2(x+1)$



71. $f(x) = 3^{x+2} - 4$, $x = f(y) = 3^{y+2} - 4 \to y = f^{-1}(x) = \log_3(x+4) - 2$

6.2 Properties of Logarithms

Theorem 6.2.1 (Inverse Properties of Exponential and Logarithmic Functions)**.** *Let $b > 0$, $b \neq 1$.*

- $b^a = c$ *if and only if* $\log_b(c)$.
- $\log_b(b^x) = x$ *for all* x *and* $b^{\log_b(x)} = x$ *for all* $x > 0$.

Theorem 6.2.2 (One-to-one Properties of Exponential and Logarithmic Functions)**.** *Let $f(x) = b^x$ and $g(x) = log_b(x)$ where $b > 0$, $b \neq 1$. Then f and g are one-to-one and*

- $b^u = b^w$ *if and only if* $u = w$ *for all real numbers* u *and* w.
- $\log_b(u) = \log_b(w)$ *if and only if* $u = w$ *for all real numbers* $u > 0$ *and* $w > 0$.

Theorem 6.2.3 (Algebraic Properties of Exponential Functions)**.** *Let $f(x) = b^x$ be an exponential function $(b > 0, b \neq 1)$ and let u and w be real numbers.*

- ***Product Rule****: $f(u + w) = f(u)f(w)$. In other words, $b^{u+w} = b^u b^w$.*
- ***Quotient Rule****: $f(u - w) = \frac{f(u)}{f(w)}$. In other words, $b^{u-w} = \frac{b^u}{b^w}$.*
- ***Power Rule****: $f(u)^w = f(uw)$. In other words, $(b^u)^w = b^{uw}$.*

Theorem 6.2.4 (Algebraic Properties of Logarithmic Functions)**.** *Let $g(x) = \log_b(x)$ be a Logarithmic function $(b > 0, b \neq 1)$ and let $u > 0$ and $w > 0$ be real numbers.*

- ***Product Rule****: $g(uw) = g(u) + g(w)$. In other words, $\log_b(uw) = \log_b(u) + \log_b(w)$.*
- ***Quotient Rule****: $g(\frac{u}{w}) = g(u) - g(w)$. In other words, $\log_b(\frac{u}{w}) = \log_b(u) - \log_b(w)$.*
- ***Power Rule****: $g(u^w) = wg(u)$. In other words, $\log_b(u^w) = w \log_b(u)$.*

Theorem 6.2.5 (Change of Base Formulas)**.** *Let $a > 0, b > 0, a \neq 1, b \neq 1$.*

- $a^x = b^{x \log_b(a)}$ *for all real numbers* x.
- $\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$ *for all real numbers* $x > 0$.

6.2.1 Exercises

1. $\ln(x^3y^2) = \ln(x^3) + \ln(y^2) = 3\ln(x) + 2\ln(y)$

13. $\log\left(\frac{100x\sqrt{y}}{\sqrt[3]{10}}\right) = \log(100x\sqrt{y}) - \log(\sqrt[3]{10}) = \log(100) + \log(x) + \log(\sqrt{y}) - 1/3 = 2 + \log(x) + \frac{1}{2}\log(y) - 1/3 =$
$= \frac{5}{3} + \log(x) + \frac{1}{2}\log(y)$

16. $4\ln(x) + 2\ln(y) = \ln(x^4) + \ln(y^2) = \ln(x^4y^2)$

28. $\log_2(x) + \log_4(x-1) = \frac{\log_4(x)}{\log_4(2)} + \log_4(x-1) = 2\log_4(x) + \log_4(x-1) = \log_4(x^2) + \log_4(x-1)$
$= \log_4(x^2(x-1)) = \log_2(x\sqrt{x-1})$

30. 7^{x-1} to base e. $7^{x-1} = e^{(x-1)\log_e(7)}$

34. $\log_3(12) = \frac{\ln(12)}{\ln(3)} \approx 2.26$

6.3 Exponential Equations and Inequalities

Procedure 6.3.1 (Steps for solving an Equation involving Exponential Functions)**.**

1. Isolate the exponential function.
2. a If convenient, express both sides with a common base and equate the exponents.

   b Otherwise, take the natural log of both sides of the equation and use the Power Rule.

6.3.1 Exercises

1. $2^{4x} = 8$

$2^{4x} = 2^3$

$4x = 3$

$x = 3/4$

13. $(1.005)^{12x} = 3$

$12x = \log_{1.005}(3)$

$x = \dfrac{\ln(3)}{12\ln(1.005)}$

19. $\dfrac{100e^x}{e^x+2} = 50$

$100e^x = 50e^x + 100$

$50e^x = 100$

$e^x = 2$

$x = \ln 2$

31. $e^x - 3e^{-x} = 2$

$e^x - 3e^{-x} = 2$

$\underline{e^{2x} - 2e^x - 3 = 0}$

$t^2 - 2t - 3 = 0$

$t = \dfrac{2 \pm \sqrt{4+12}}{2} = \dfrac{2 \pm \sqrt{16}}{2} = \dfrac{2 \pm 4}{2} = 1 \pm 2$

$t_1 = 3, \; t_2 = -1$

$e^x = 3$ $\qquad$ $e^x = -1$

$\underline{\underline{x = \ln(3)}}$ $\qquad$ $e^x = \ln(-1)$

34. $e^x > 53$

$\ln(e^x) > \ln(53)$

$x > \ln(53)$

38. $\dfrac{150}{1+29e^{-0.86}} \leq 130$

$150 \leq 130(1+29e^{-0.86})$

$150 \leq 130 + 3770e^{-0.86}$

$\frac{20}{3770} \leq e^{-0.86}$

$-0.86 \geq \ln(\frac{20}{3770})$

$t \leq \frac{-1}{0.86}\ln(\frac{20}{3770})$

D: $(-\infty, \frac{5}{4}\ln(\frac{377}{2})]$

6.4 Logarithmic Equations and Inequalities

Procedure 6.4.1 (Steps for Solving an Equation involving Logarithmic Functions)**.**

1. Isolate the logarithmic function.
2. b If convenient, express both sides as logs with the same base and equate the arguments of the log functions.

   b Otherwise, rewrite the log equation as an exponential equation.

6.4.1 Exercises

1. $\log(3x-1) = \log(4-x)$

$3x-1 = 4-x$

$4x = 5$

$x = 5/4$

11. $-\log(x) = 5.4$

$\log(x^{-1}) = 5.4$

$x^{-1} = 10^{5.4}$

$x = 10^{-5.4}$

17. $\log_{169}(3x+7) - \log_{169}(5x-9) = 1/2$

$\log_{169}\left(\frac{3x+7}{5x-9}\right) = 1/2$

$\frac{3x+7}{5x-9} = \sqrt{169}$

$3x+7 = 13(5x-9)$

$3x - 65x = -7 - 117$

$62x = 124$

$x = 2$

23. $(\log(x))^2 = 2\log(x) + 15 \qquad x > 0$

$(\log(x))^2 - 2\log(x) - 15 = 0$

$l = \log(x)$

$l^2 - 2l - 15, \quad l_1 = 5, \; l_2 = -3$

$\log(x) = 5 \rightarrow x = 10^5$

$\log(x) = -3 \rightarrow x = 10^{-3}$

25. $\frac{1-\ln(x)}{x^2} < 0 \qquad x \neq 0$

$1 - \ln(x) < 0$

$-\ln(x) < -1$

$\ln(x) > 1$

$x > e \qquad (e, \infty)$

29. $2.3 < -\log(x) < 5.4$

$2.3 < \log(x^{-1}) < 5.4$

$10^{2.3} < x^{-1} < 10^{5.4}$

$10^{-5.4} < x < 10^{-2.3} \qquad (10^{-5.4}, 10^{-2.3})$

Chapter 7

Hooked on Conics

7.1 Introduction to Conics

7.2 Circles

7.2.1 Exercises

7.3 Parabolas

7.3.1 Exercises

7.4 Ellipses

7.4.1 Exercises

7.5 Hyperbolas

7.5.1 Exercises